

Engineering Project Report

Wireless Audio Transmission System Using Modulated LASER Light (Li-Fi Technology)

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Abstract

This technical report details the design, construction, and field evaluation of a free-space optical audio transmission system based on Light Fidelity (Li-Fi) principles. The system uses a MAX4466 electret microphone pre-amplifier and a PAM8403 Class-D audio amplifier to intensity-modulate a 650 nm, 5 mW red semiconductor laser diode with analog audio signals. At the receiver end, a mini solar panel acts as a large-area photovoltaic detector, converting light intensity fluctuations into a proportional photocurrent. A secondary PAM8403 amplifier boosts the signal to drive a 4 Ω , 10 W speaker. Empirical field testing demonstrated clear audio reception across indoor distances up to 50 meters and line-of-sight outdoor distances exceeding 100 meters.

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1 Introduction & System Overview

Visible Light Communication (VLC) and Li-Fi utilize visible light spectrum (400 nm – 700 nm) for high-speed, zero-RF-interference data and analog audio transmission.

The objective of this project is to implement a robust, low-cost optical audio link transmitting voice and music over a focused laser beam directly onto a photovoltaic solar panel photodetector.

2 Bill of Materials

Table 1: Bill of Materials for Li-Fi Wireless Audio Link

S.No.	Component	Qty	Part Number	Circuit Function
1	Mic Pre-Amplifier	1	MAX4466 Module	Electret microphone pre-amp with adjustable gain
2	Audio Amplifiers	2	PAM8403 Modules	3W + 3W Class-D dual-channel power amps
3	Laser Diode	1	650 nm Red (5V, 5mW)	Optical light transmitter
4	Current Resistor	1	30 Ω , 1/4W	Limits laser forward drive current
5	Photodetector	1	Mini Solar Panel (5V)	Large surface-area optical photodiode array
6	Output Speaker	1	4 Ω , 10 Watt	Transducer converting audio electrical signals to sound
7	Voltage Regulators	2	LM7805	5 V DC regulators for Tx and Rx modules
8	Power Supplies	2	9 V DC Batteries	Isolated power sources for Tx and Rx

3 System Operating Principle

3.1 Transmitter Stage (Acoustic to Optical Modulation)

1. Acoustic sound waves strike the electret microphone, producing millivolt-level electrical signals amplified by the MAX4466 module.
2. The output signal enters the PAM8403 Class-D amplifier module.
3. The PAM8403 output drives the 650 nm laser diode. The instantaneous laser output intensity (P_{opt}) varies in direct proportion to the forward current (I_F):

$$P_{\text{opt}}(t) = P_0 + \eta \cdot i_{\text{audio}}(t) \quad (1)$$

where P_0 is the baseline optical power, η is differential efficiency, and $i_{\text{audio}}(t)$ is the amplified audio current.

3.2 Receiver Stage (Optical to Acoustic Demodulation)

1. Modulated laser light hits the mini solar panel. Photons induce electron-hole pairs via the photoelectric effect, generating photocurrent $i_{\text{ph}}(t)$:

$$i_{\text{ph}}(t) = R \cdot P_{\text{opt}}(t) \quad (2)$$

where R is solar panel responsivity (A/W).

2. The AC photocurrent component passes through a $100\text{ k}\Omega$ biasing pot into the second PAM8403 amplifier.
3. The PAM8403 amplifies the audio signal to drive the $4\ \Omega$ 10 W speaker.

4 Comparison: Li-Fi vs. Wi-Fi vs. Bluetooth

Table 2: Technology Comparison Matrix

Parameter	Li-Fi (Laser Audio)	Wi-Fi	Bluetooth
Medium	Visible Light (650 nm)	RF (2.4 GHz/5 GHz)	RF (2.4 GHz)
Line-of-Sight	Required (Strict)	Not Required	Not Required
RF Interference	Zero (RF Safe)	Moderate / High	High
Latency	$< 1\text{ ms}$	$5\text{ ms} - 20\text{ ms}$	$20\text{ ms} - 100\text{ ms}$
Security	High (Cannot cross walls)	Encryption Needed	Pairing Needed

5 Field Testing & Range Results

- **Indoor Test Range:** $10\text{ m} - 50\text{ m}$ (Clear audio playback without noticeable distortion).
- **Outdoor Range:** $> 100\text{ m}$ along direct line-of-sight.
- **Noise Suppression:** A dark optical hood over the solar panel shields ambient sunlight DC noise.

6 Conclusion

The Laser Li-Fi Wireless Audio Transfer System successfully demonstrates high-fidelity optical audio communication over long line-of-sight distances, offering zero RF emissions and low signal latency.